

SAFETY DATA SHEET (GHS)

Acetone | CAS: 67-64-1 | Formula: C₃H₆O

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Acetone, CAS 67-64-1 (<https://echa.europa.eu/>)
- PubChem CID 180 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/180>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0087 — Acetone
- ACGIH TLV/BEI — Acetone (TWA 250 ppm / STEL 500 ppm)
- NOAA CAMEO Chemicals — Acetone datasheet (<https://cameochemicals.noaa.gov/>)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Acetone

CAS Number: 67-64-1

Chemical formula: C₃H₆O

Molecular weight: 58.08 g/mol

Synonyms: 2-Propanone; Dimethyl ketone; Propan-2-one

Intended use: Co-product of the cleavage (decomposition) reaction — produced alongside phenol from CHP decomposition; recovered as a major saleable product. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Liq. 2 (H225); Eye Irrit. 2A (H319); STOT SE 3 (H336)

Signal word: Danger

Hazard statements (H-codes): H225 (Highly flammable liquid and vapour); H319 (Causes serious eye irritation); H336 (May cause drowsiness or dizziness)

Precautionary statements (P-codes): P210, P233, P240, P241, P242, P243, P261, P264, P271, P280, P303+P361+P353, P304+P340, P305+P351+P338, P337+P313, P370+P378, P403+P233, P405, P501

GHS pictograms: GHS02 (Flammable), GHS07 (Irritant)

Other hazards: Highly flammable with a very low flash point (-20°C) and wide flammable range (LEL 2.5% / UEL 12.8%) — significant fire hazard in the cleavage/distillation product recovery section. Vapours can accumulate in low areas (vapour density >1). High vapour pressure means large vapour volumes can be generated rapidly from a spill. Not classified as a carcinogen, mutagen, or reproductive toxicant; relatively low chronic toxicity compared to many other Train II chemicals, but the flammability hazard is severe.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Acetone: CAS 67-64-1 — product-grade purity typically >99%; confirm against PPCL acetone product specification (trace water, methanol, or other carbonyl by-products possible from the cleavage reaction)

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. If breathing is difficult, administer oxygen. Obtain medical attention if symptoms (drowsiness, dizziness, headache) persist.

Skin contact: Wash with soap and water. Acetone is mildly defatting on repeated/prolonged contact — moisturise affected skin. Remove contaminated clothing.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention if irritation persists.

Ingestion: Do NOT induce vomiting. Rinse mouth with water. Obtain medical attention if a significant quantity is swallowed — may cause CNS depression (drowsiness) and gastrointestinal irritation.

Notes to physician: Acetone is metabolised to a minor extent to isopropanol-related ketone bodies; monitor for CNS depression following large ingestion/inhalation exposures; generally good prognosis with supportive care

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Alcohol-resistant foam (important — acetone is a polar/water-miscible solvent that breaks down standard foams), CO₂, dry chemical, water spray/fog for cooling

Unsuitable extinguishing media: Standard (non-alcohol-resistant) protein/AFFF foams may be ineffective due to acetone's miscibility with water

Special hazards: Highly flammable liquid and vapour (H225) with flash point -20°C — vapours can ignite at well below ambient temperature. Vapours heavier than air can travel to ignition sources and flash back. Significant vapour generation from spills due to high volatility/vapour pressure.

Special protective equipment for firefighters: Full bunker gear with positive-pressure SCBA; cool fire-exposed vessels with water spray from a safe distance

Hazardous combustion products: Carbon monoxide, carbon dioxide

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Eliminate all ignition sources immediately. Evacuate non-essential personnel. Use explosion-proof equipment. Bond/ground transfer equipment to control static discharge — acetone can accumulate static charge during pumping.

Environmental precautions: Acetone is readily biodegradable and has relatively low ecotoxicity, but prevent large releases to drains/watercourses to avoid oxygen depletion (BOD) effects and fire hazard in sewers

Containment and cleanup: Contain spill with non-combustible absorbent. Ventilate area thoroughly given high volatility. Collect in closed, labelled containers for recovery/disposal.

Reference to other sections: See Section 7 (Handling/Storage) and Section 9 (flammability data)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Handle in closed systems; use explosion-proof/intrinsically safe electrical equipment in classified areas. Bond and ground transfer lines/containers — acetone is a good static accumulator. Avoid generating vapours; ensure adequate ventilation, especially around the product distillation/recovery columns and storage tanks.

Storage conditions: Store in closed tanks in a well-ventilated area, away from heat, sparks, open flames and other ignition sources. Conservation venting to vapour recovery where required by environmental permit (acetone is a VOC).

Temperature limits: Store below ambient where practical; given the very low flash point, essentially all realistic ambient/process temperatures in tropical Thailand are above the flash point — treat all acetone storage/handling areas as permanently within the flammable vapour risk zone for area-classification purposes

Pressure: Atmospheric/low-pressure storage with conservation venting

Segregation requirements: Segregate from strong oxidisers (can form peroxides/unstable mixtures with some oxidisers), strong acids/bases (can catalyse unwanted condensation reactions), and sources of ignition

Special requirements: LEL monitoring in storage and product recovery areas; static control (bonding/grounding) on all transfer operations

Specific end-use recommendations: As a co-product, ensure storage tank design accommodates the full production rate envelope including upset/off-spec conditions; confirm vapour recovery system capacity given acetone's high volatility

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: 250 ppm; STEL: 500 ppm. OSHA PEL: 1000 ppm TWA. Thailand OEL: confirm current Ministry of Labour value.

TWA: 250 ppm (ACGIH) / 1000 ppm (OSHA)

STEL: 500 ppm (ACGIH)

Ceiling: Not established

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: Closed system handling; local exhaust ventilation at sample points and product transfer operations; LEL monitoring

PPE — Respiratory: Air-purifying respirator with organic vapour cartridge for tasks with vapour exposure potential above OEL; SCBA for tank entry or major release response

PPE — Gloves: Chemical-resistant gloves (nitrile or other solvent-resistant material — note acetone permeates many common glove materials quickly; verify breakthrough time and consider more frequent glove changes)

PPE — Eyes: Chemical splash goggles for tasks with splash potential

PPE — Body: Flame-resistant (FR) clothing; antistatic footwear in classified areas

Biological monitoring: Urinary acetone has been used as a biomarker in some occupational health programmes, though acetone is also an endogenous metabolite (confounds interpretation)

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless liquid

Odour: Characteristic sweetish, fruity ('pear drop'-like) solvent odour

Odour threshold: ~13-42 ppm (varies by source)

pH: Not applicable (neutral organic solvent; miscible with water)

Melting point / Freezing point: -94.7°C

Boiling point (at 1 atm): 56.05°C

Flash point: -20°C (closed cup) — experimental literature value

Evaporation rate: ~5.6-7.7 (n-butyl acetate = 1) — evaporates much faster than water

Flammability (solid/gas): Highly flammable liquid

Flammability limits in air: LEL 2.5 vol%; UEL 12.8 vol%

Vapour pressure (at 20°C): ~24.6 kPa (~185 mmHg) — high volatility

Vapour density (air = 1): ~2.0 (heavier than air)

Relative density (water = 1): ~0.79 (at 20°C)

Solubility (water): Fully miscible (infinite solubility)

Partition coefficient (log Kow): ~-0.24 (hydrophilic)

Auto-ignition temperature: ~465°C

Decomposition temperature: Not applicable under normal use

Viscosity: ~0.31 mPa·s at 25°C (very low — mobile liquid)

Explosive properties: Not classified as explosive substance; forms flammable vapour/air mixtures within a wide LEL-UEL range

Oxidising properties: Not an oxidiser; can form unstable peroxide-type mixtures with certain strong oxidisers (e.g. concentrated H₂O₂ + acetone forms shock-sensitive peroxides — relevant if acetone ever contacts peroxide-containing streams such as CHP)

Molecular weight: 58.08 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; reactive as a ketone — can undergo condensation reactions with itself (aldol-type) under strongly basic conditions, and reacts with oxidisers

Chemical stability: Stable under recommended storage and handling conditions

Possibility of hazardous reactions: Forms shock-sensitive peroxide compounds on contact with concentrated hydrogen peroxide or other strong peroxidising agents — relevant cross-contamination hazard if acetone streams could contact CHP-containing streams (CAS 80-15-9) in an upset condition; reacts vigorously with strong oxidisers (chromic acid, nitric acid mixtures); self-condensation under strongly basic/acidic catalytic conditions can be exothermic

Conditions to avoid: Heat, sparks, open flames, static discharge; contact with concentrated oxidisers/peroxides; cross-contamination with the CHP-containing oxidation/decomposition streams

Incompatible materials: Strong oxidisers (especially peroxides — see above), strong reducing agents, halogenated compounds (some exothermic reactions), strong bases/acids in concentrated form

Hazardous decomposition products: None under normal storage; combustion produces CO, CO₂

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation (primary), skin/eye contact, ingestion

Acute toxicity — Oral LD₅₀: ~5,800 mg/kg, rat (low acute oral toxicity)

Acute toxicity — Dermal LD₅₀: >15,000 mg/kg, rabbit (very low acute dermal toxicity)

Acute toxicity — Inhalation LC₅₀: ~50,000-76,000 ppm (4 h), rat (low acute inhalation lethality; narcotic/irritant effects occur at much lower concentrations)

Skin corrosion/irritation: Not classified as a skin irritant under GHS, though repeated contact has a defatting effect

Serious eye damage/irritation: Causes serious eye irritation (H319)

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified by IARC or other major bodies

Reproductive toxicity: Not classified

STOT (single exposure): May cause drowsiness or dizziness (H336) — CNS depressant at high vapour concentrations

STOT (repeated exposure): Not classified

Aspiration hazard: Not classified (low viscosity but not classified as aspiration hazard under GHS for this substance)

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC₅₀/EC₅₀): Fish LC₅₀ (96h) ~5,000-8,300 mg/L; Daphnia EC₅₀ (48h) ~8,000-12,600 mg/L — low aquatic toxicity; not classified for aquatic hazards under GHS

Persistence and degradability: Readily biodegradable — high BOD/COD contribution if released to water bodies in large quantity (oxygen depletion risk rather than direct toxicity)

Bioaccumulative potential: Negligible (log K_{ow} ~-0.24)

Mobility in soil: Highly mobile (fully water miscible, volatile)

Other adverse effects: VOC — relevant to air emissions permitting; not ozone-depleting

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Recover/recycle where possible (acetone is a saleable product); off-spec material should be disposed of via licensed hazardous waste incineration or as per site waste management procedures; do not discharge large quantities to drains (BOD/fire hazard in sewers)

Contaminated packaging: Drain thoroughly; dispose of as per local regulations or return for reconditioning

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535; site effluent discharge permit (COD/BOD limits)

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1090

Proper shipping name: ACETONE

Transport hazard class(es): Class 3 (Flammable liquid)

Packing group: II

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: Ensure bonding/grounding during loading/unloading given high static-accumulating tendency; vapour-tight, vented tanks

Bulk transport (MARPOL/IBC): Classify per IBC Code if shipped in bulk by sea

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 67-64-1)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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